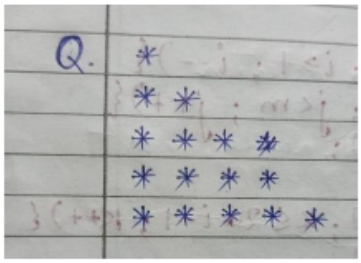
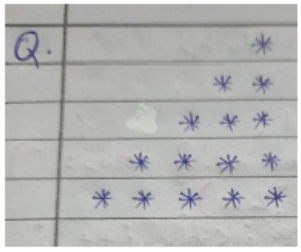
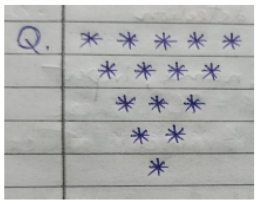
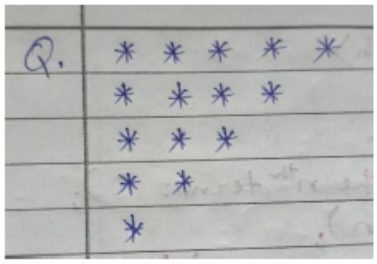
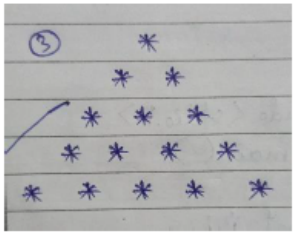
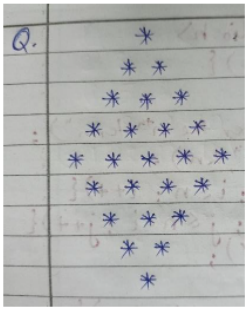
Programming for Problem Solving
B.Tech [2nd Semester], 2024
Questions

Q.1. Write a C-Program to print “Hello World!”.
Q.2. Write a C-Program to add/subtract/multiply/divide two numbers.
Q.3. Write a C-Program to find the average of five numbers.
Q.4. Write a C-Program to display the size of different data types.
Q.5. Write a C-Program to find the area and circumference of a circle.
Q.6. Write a C-Program to find the area of a circle.
Q.7. Write a C-Program to check whether a character is Vowel/Consonant.
Q.8. Write a C-Program to check whether a number is Even/Odd.
Q.9. Write a C-Program to find the greatest among three numbers.
Q.10. Write a C-Program to swap the digits of any given number.
Q.11. Write a C-Program to assign grades to given marks in a test.
Q.12. Write a C-Program to find the smallest and largest number among five numbers.
Q.13. Write a C-Program to check leap year.
Q.14. Write a C-Program to check the eligibility to vote.
Q.15. Write a C-Program to find the Fibonacci sequence.
Q.16. Write a C-Program to print all the prime numbers between 1 to n.
Q.17. Write a C-Program to print all the prime numbers between any two given numbers.
Q.18. Write a C-Program to print all the odd and even numbers between 1 to n.
Q.19. Write a C-Program using switch case, to make a simple calculator.
Q.20. Write a C-Program to print the multiplication table of any number.
Q.21. - Q.26. Write a C-Program to print the following pattern:

Q.27. Write a C-Program to add the following series upto ‘n’ terms: 9+99+999+9999+…..+n.

Q.28. Write a C-Program to add the following series upto ‘n’ terms: 1+ 1/2 + 1/3 + 1/4 + ….. + $1/n$.

Q.29. Write a C-Program to add the following series upto ‘n’ terms: 1 + 3^2 + 5^3 + 7^4 + 9^5 + ……..

Q.30. Write a C-Program to find the factorial of a number using function.

Q.31. Write a program in C to find the square of any number using the function.

Q.32. Write a program in C to swap two numbers using a function.

Q.33. Write a program in C to find the sum of the series 1!/1+2!/2+3!/3+4!/4+5!/5 using the function.

Q.34. Write a C program to find the maximum and minimum of some values using a function that returns an array.

Q.35. Write a program in C to add two numbers using pointers.

Q.36. Write a program in C to add numbers using call by reference.

Q.37. Write a program in C to find the maximum number between two numbers using a pointer.

Q.38. Write a program in C to store n elements in an array and print the elements using a pointer.

Q.39. Write a program in C to swap elements using call by reference.

Q.40. Write a program in C to find the factorial of a given number using pointers.

Q.41. Write a program in C to count the number of vowels and consonants in a string using a pointer.

Q.42. Write a program in C to compute the sum of all elements in an array using pointers.

Q.43. Write a program in C to print all the alphabets using a pointer.

Q.44. Write a program in C to print a string in reverse using a pointer.

Q.45. Write a C program that takes input from the user and counts the number of uppercase and lowercase letters, as well as the number of other characters.

Q.46. Write a C program to check whether a number is an Armstrong Number or not.

Q.47. Write a C program to check whether a number is a Palindrome or not.

Q.48. Write a C program to check whether a given number is a Perfect number or not.

Q.49. Write a program in C to find the sum of the series: x - x^3 + x^5 - .....

Q.50. Write a program in C to convert a decimal number into binary without using an array.

Miscellaneous Questions

Q. Write a C program to find the length of a string without using the library function.

Q. Write a C program to find the sum of an A.P. series.

Q. Write a C program to find the sum of the G.P. series.

Q. Write a program in C to find the LCM of any two numbers.

Q. Write a program in C to find the LCM of any two numbers.
Q. Write a C program to find the LCM of any two numbers using HCF.
Q. Write a C program to display the sum of n terms of even natural numbers.
Q. Write a C program to display the sum of n terms of odd natural numbers.
Q. Write a C program to display a pattern like a right angle triangle with a number.

```
1
1 2
1 2 3
1 2 3 4
```

Q. Write a program in C to make such a pattern like a right angle triangle with a number which will repeat a number in a row.

```
1
2 2
3 3 3
4 4 4 4
```

Q. Write a program in C to make such a pattern like a right angle triangle with the number increased by 1.

```
1
2 3
4 5 6
7 8 9 10
```

Q. Write a program in C to make a pyramid pattern with numbers increased by 1.

```
      1
     2 3
    4 5 6
   7 8 9 10
```

Q. Write a program in C to print Floyd's Triangle.

```
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```